```c
#include<stdio.h>
int ctr = 0;
void reverse(int n)
{
   int rev = 0;++ctr; int remainder;++ctr;
   while (n != 0)
    {
        ++ctr;
        remainder = n % 10;++ctr;
        rev = rev * 10 + remainder;++ctr;
        n/= 10;++ctr;

    }
    ++ctr;
printf("%d\n",ctr);
}
int main(){
    int n;scanf("%d",&n);
    reverse(n);
}
```

```c
#include<stdio.h>
int cnt = 0;
void function(int n)
{
    int c= 0;
    ++cnt;
    for(int i=n/2; i<n; i++){
        ++cnt;
        for(int j=1; j<n; j = 2 * j){
            ++cnt;
            for(int k=1; k<n; k = k * 2){
                ++cnt;
                c++;
                ++cnt;
            }
            ++cnt;
        }
        ++cnt;
    }
    ++cnt;
    printf("%d\n",cnt);
}
int main(){
    int n ;
    scanf("%d",&n);
    function(n);
}
```

```c
#include<stdio.h>
int cnt = 0;
void Factor(int num)
 {
    for (int i = 1; i <= num;++i){
        ++cnt;
        ++cnt;
        if (num % i== 0){
            ++cnt;
            //printf("%d ", i);
        }
     }
     ++cnt;
     printf("%d\n",cnt);
  }

int main(){
    int n;
    scanf("%d",&n);
    Factor(n);
}
```

```c
#include<stdio.h>
int ctr = 0;
void func(int n){
    ++ctr;
    if(n==1){
        //printf("*");
    }
    else{
     for(int i=1; i<=n; i++){
         ++ctr;
         ++ctr;
         for(int j=1; j<=n; j++){
          ++ctr;
          ++ctr;
          break;
         }
        ++ctr;
     }
     ++ctr;
   }

   printf("%d\n",ctr);
 }
 int main(){
     int n;
     scanf("%d",&n);
     func(n);
 }
```

```c
#include<stdio.h>
int ctr = 0;
void function (int n)
{
    int i= 1;++ctr;
    int s =1;++ctr;
    ++ctr;
    while(s <= n)
    {
         i++;++ctr;
         s += i;++ctr;
         ++ctr;
     }
    printf("%d",ctr);
}
int main(){
    int n;
    scanf("%d",&n);
    function(n);
}
```

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) return 0;

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int tails[n];
    int len = 0;

    for (int i = 0; i < n; i++) {
        // Binary search to find the upper bound (first element > arr[i])
        int low = 0, high = len;
        while (low < high) {
            int mid = low + (high - low) / 2;
            if (tails[mid] <= arr[i]) {
                low = mid + 1; // Allows duplicate elements for non-decreasing
            } else {
                high = mid;
            }
        }

        tails[low] = arr[i];
        if (low == len) {
            len++;
        }
    }

    printf("%d\n", len);
    return 0;
}

```

```c
#include <stdio.h>
#include <string.h>

int max(int a, int b) {
    return (a > b) ? a : b;
}

int main() {
    char s1[1005], s2[1005];
    if (scanf("%s %s", s1, s2) != 2) return 0;

    int m = strlen(s1);
    int n = strlen(s2);
    int dp[m + 1][n + 1];

    for (int i = 0; i <= m; i++) {
        for (int j = 0; j <= n; j++) {
            if (i == 0 || j == 0) {
                dp[i][j] = 0;
            } else if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            } else {
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
            }
        }
    }

    printf("%d\n", dp[m][n]);
    return 0;
}

```

```c
#include <stdio.h>

long long max(long long a, long long b) {
    return (a > b) ? a : b;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) return 0;

    long long grid[n][n];
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%lld", &grid[i][j]);
        }
    }

    // dp array stores the maximum value to reach each cell in the current row
    long long dp[n];

    // Initialize base case for the top-left cell
    dp[0] = grid[0][0];

    // Initialize the rest of the first row (can only come from the left)
    for (int j = 1; j < n; j++) {
        dp[j] = dp[j - 1] + grid[0][j];
    }

    // Process the remaining rows
    for (int i = 1; i < n; i++) {
        // First cell of the current row can only come from above
        dp[0] += grid[i][0];

        // For the rest of the cells, choose the max between coming from above or left
        for (int j = 1; j < n; j++) {
            dp[j] = max(dp[j], dp[j - 1]) + grid[i][j];
        }
    }

    printf("%lld\n", dp[n - 1]);
    return 0;
}

```

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    if (n == 0 || n == 1 || n == 2) {
        printf("1\n");
        return 0;
    }

    long long a = 1, b = 1, c = 1, current = 0;
    for (int i = 3; i <= n; i++) {
        current = c + a;
        a = b;
        b = c;
        c = current;
    }

    printf("%lld\n", c);
    return 0;
}

```

```c
#include<stdio.h>
void swap(int *a,int *b){
    int temp = *a;
    *a = *b;
    *b = temp;
}
int partition(int a[],int l,int h){
    int i = l - 1;
    int p = a[h];
    for(int j=l;j<h;++j){
        if(a[j] < p){
            ++i;
            swap(&a[j],&a[i]);
        }
    }
    swap(&a[h],&a[i+1]);
    return i + 1;
}
void qs(int a[],int l,int h){
    if(l <= h){
        int p = partition(a,l,h);
        qs(a,l,p-1);
        qs(a,p+1,h);
    }
}
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;++i) scanf("%d",&arr[i]);
    int arr2[n];
    for(int i=0;i<n;++i) scanf("%d",&arr2[i]);
    qs(arr,0,n-1);
    qs(arr2,0,n-1);
    int ans = 0;
    /*for(int i=0;i<n;++i) printf("%d ",arr[i]);
    printf("\n");
    for(int i=0;i<n;++i) printf("%d ",arr2[i]);*/
    for(int i=0;i<n;++i) ans+= (arr[i] * arr2[n-1-i]);
    printf("%d\n",ans);

}
```

```c
#include<stdio.h>
void swap(int *a,int *b){
    int temp = *a;
    *a = *b;
    *b = temp;
}
int partition(int arr[],int low,int high){
    int pivot = arr[high];
    int i = low - 1;
    for(int j = low ; j <= high ; ++j){
        if(arr[j] < pivot){
            ++i;
            swap(&arr[i],&arr[j]);
        }
    }
    swap(&arr[high],&arr[i+1]);
    return i + 1;
}
void QuickSort(int arr[],int low,int high){
    if(low <= high){
        int p = partition(arr,low,high);
        QuickSort(arr,low,p - 1);
        QuickSort(arr,p + 1,high);
    }
}
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;++i) scanf("%d",&arr[i]);
    QuickSort(arr,0,n-1);
    int ans = 0;
    for(int i=0;i<n;++i) ans += (arr[i] * i);
    printf("%d\n",ans);
}
```

```c
#include<stdio.h>
#include<math.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    int sum=0;

    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<n;i++){
        for(int j=0;j<n-i-1;j++){
            if(arr[j]<arr[j+1]){
                int temp=arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=temp;
            }
        }
    }
    for(int i=0;i<n;i++){
        int s=pow(n,i);
        sum+=(arr[i]*s);
    }
    printf("%d",sum);

}
```

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;++i) scanf("%d",&arr[i]);
    int m;
    scanf("%d",&m);
    int arr2[m];
    for(int i=0;i<m;++i) scanf("%d",&arr2[i]);
    int i = 0,j = 0;
    int ans = 0;
    while(i < n && j < m){
        if(arr2[j] >= arr[i]){
            ++ans;
            ++i;
            ++j;
        }
        else ++j;
    }
    printf("%d\n",ans);
}
```

```c
#include<stdio.h>
#include<stdbool.h>
int main(){
    int arr[9] = {1,2,5,10,20,50,100,500,1000};
    int back = 8;
    int target;
    scanf("%d",&target);
    int ans = 0;
    while(true){
        if(target > arr[back]){
            while(target >= arr[back]) {target -= arr[back];++ans;}
            if(target == 0) break;
        }
        else --back;
    }
    printf("%d\n",ans);
}
```

```c
#include<stdio.h>
void swap(int *a,int *b){
    int temp = *a;
    *a = *b;
    *b = temp;
}
int partition(int arr[],int low,int high){
    int i = low - 1;
    int pivot = arr[high];
    for(int j = low ;j <= high;++j){
        if(arr[j] < pivot){
            ++i;
            swap(&arr[i],&arr[j]);
        }
    }
    swap(&arr[i + 1],&arr[high]);
    return i+1;
}
void quicksort(int arr[],int low,int high){
    if(low < high){
        int p = partition(arr,low,high);
        quicksort(arr,low,p - 1);
        quicksort(arr,p+1,high);
    }
}
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;++i) scanf("%d",&arr[i]);
    int low = 0,high = n - 1;
    quicksort(arr,low,high);
    for(int i=0;i<n;++i){
        printf("%d ",arr[i]);
    }

}
```

```c
#include<stdio.h>
#include<stdbool.h>
bool binary_search(int n,int arr[],int x){
    int start = 0; int end = n - 1;
    while(start <= end){
        int mid = start + (end - start)/2;
        if(arr[mid] == x){
            return true;
        }

        else if(arr[mid] > x){
            end = mid - 1;
        }
        else start = mid + 1;
    }
    return false;
}
bool function(int arr[],int n,int x){
    for(int i=0;i<n;++i){
        int first = arr[i];
        int second = x - first;
        if(binary_search(n,arr,second)){
            printf("%d\n%d\n",first,second);
            return true;
        }
    }
    return false;
}
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;++i) scanf("%d",&arr[i]);
    int x;
    scanf("%d",&x);
    if(!function(arr,n,x)){
        printf("No\n");
        return 0;
    }
}
```

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;++i) {scanf("%d",&arr[i]);
        //printf("%d ",arr[i]);
    }
    int x;

    scanf("%d",&x);
    int right = n - 1;
    while(right >= 0){
        if(arr[right] > x) --right;
        else {printf("%d\n",arr[right]);return 0;}
    }
}
```

```c
#include<stdio.h>
#include<limits.h>
int main(){
    int n;
    scanf("%d",&n);
    int ans = INT_MIN;
    for(int i=0;i<n;++i){
        int x;
        scanf("%d",&x);
        ans = (x >= ans? x : ans );
    }
    printf("%d\n",ans);
}
```

```c
#include<stdio.h>
int main(){
    int m;
    scanf("%d",&m);
    int ans = 0;
    for(int i=0;i<m;++i){
        int x;
        scanf("%d",&x);
        ans+=(x == 0);
    }
    printf("%d\n",ans);
}
```

```c
#include <stdio.h>
#include <stdbool.h>

bool findPairWithDifference(int A[], int n, int k) {
    int i = 0;
    int j = 1;

    while (i < n && j < n) {
        if (i == j) {
            j++;
            continue;
        }

        int diff = A[j] - A[i];

        if (diff == k) {
            return true;
        } else if (diff < k) {
            j++;
        } else {
            i++;
        }
    }

    return false; // No such pair exists
}

int main() {
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i = 0;i<n;++i) scanf("%d",&arr[i]);
    int k;
    scanf("%d",&k);

    if (findPairWithDifference(arr, n, k)) {
        printf("1\n");
    } else {
        printf("0\n");
    }

    return 0;
}
```

```c
#include <stdio.h>
#include <stdbool.h>

bool findPairWithDifference(int A[], int n, int k) {
    int i = 0;
    int j = 1;

    while (i < n && j < n) {
        if (i == j) {
            j++;
            continue;
        }

        int diff = A[j] - A[i];

        if (diff == k) {
            return true;
        } else if (diff < k) {
            j++;
        } else {
            i++;
        }
    }

    return false; // No such pair exists
}

int main() {
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i = 0;i<n;++i) scanf("%d",&arr[i]);
    int k;
    scanf("%d",&k);

    if (findPairWithDifference(arr, n, k)) {
        printf("1\n");
    } else {
        printf("0\n");
    }

    return 0;
}
```

```c
#include<stdio.h>
int main(){
    int t;
    scanf("%d",&t);
    while(t--){
        int  n1,n2;
        scanf("%d",&n1);
        int a[n1];
        for(int i = 0;i<n1;++i) scanf("%d",&a[i]);
        scanf("%d",&n2);
        int a2[n2];
        for(int i = 0;i<n2;++i) scanf("%d",&a2[i]);
        int i = 0,j = 0;
        while(i<n1 && j < n2){
            if(a[i] < a2[j]){
                ++i;
            }
            else if(a[i] > a2[j]){
                ++j;
            }else{
                printf("%d ",a[i]);
                ++i;++j;
            }
        }
        printf("\n");
    }
    return 0;
}
```

```c
int main(){
    int t;
    scanf("%d",&t);
    while(t--){
        int n1,n2;
        scanf("%d",&n1);
        int a[n1];
        for(int i = 0;i<n1;++i) scanf("%d",&a[i]);
        scanf("%d",&n2);
        int a2[n2];
        for(int i = 0;i<n2;++i) scanf("%d",&a2[i]);

        for(int i = 0;i<n1;++i){
            for(int j = 0;j<n2;++j){
                if(a[i] == a2[j]){
                    printf("%d ",a[i]);
                    break;
                }
            }
        }
    }
}
```

```c
#include<stdio.h>
int getAns(int arr[]){
    int slow = arr[0],fast = arr[0];
    do{
        slow = arr[slow];
        fast = arr[arr[fast]];
    }while(slow != fast);
    int ptr1 = arr[0],ptr2 = slow;
    do{
        ptr1 = arr[ptr1];
        ptr2 = arr[ptr2];
    }while(ptr1 != ptr2);
    return ptr1;
}
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;++i){
        scanf("%d",&arr[i]);
    }
    printf("%d\n",getAns(arr));
}
```

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n+1],arr2[n];
    for(int i=0;i<=n;++i) arr[i] = 0;
    for(int i=0;i<n;++i)scanf("%d",&arr2[i]);
    for(int i=0;i<n;++i){
        ++arr[arr2[i]];
    }
    for(int i=0;i<=n;++i){
        if(arr[i] > 1){
            printf("%d\n",i);
            return 0;
        }
    }
}
```